

ILLUSTRATION

$$\blacksquare D_x(x^3)]_{x=5} = 3x^2]_{x=5} = 3(5^2) = 75$$

$$\blacksquare [D_x(9x^{4/3})]_{x=8} = [12x^{1/3}]_{x=8} = 12(8^{1/3}) = 24$$

We may also consider *derivatives of derivatives*. Specifically, if we differentiate a function f , we obtain another function f' . If f' has a derivative, it is denoted by f'' (read f double prime) and is called the **second derivative** of f . Thus,

$$f''(x) = D_x[f'(x)] = D_x[D_x f(x)] = D_x^2 f(x).$$

As we have indicated, the operator symbol D_x^2 is used for second derivatives. The **third derivative** f''' of f is the derivative of the second derivative. Thus,

$$f'''(x) = D_x[f''(x)] = D_x[D_x^2 f(x)] = D_x^3 f(x).$$

In general, if n is a positive integer, then $f^{(n)}$ denotes the **n th derivative** of f and is found by starting with f and differentiating, successively, n times. In operator notation, $f^{(n)}(x) = D_x^n f(x)$. The integer n is called the **order** of the derivative $f^{(n)}(x)$. The following summarizes various notations that are used for these **higher derivatives**, with $y = f(x)$.

Notations for higher derivatives (3.17)

$$\begin{array}{l} f'(x), f''(x), f'''(x), f^{(4)}(x), \dots, f^{(n)}(x) \\ D_x y, D_x^2 y, D_x^3 y, D_x^4 y, \dots, D_x^n y \\ y', y'', y''', y^{(4)}, \dots, y^{(n)} \\ \frac{dy}{dx}, \frac{d^2 y}{dx^2}, \frac{d^3 y}{dx^3}, \frac{d^4 y}{dx^4}, \dots, \frac{d^n y}{dx^n} \end{array}$$

EXAMPLE 5 Find the first four derivatives of $f(x) = 4x^{3/2}$.

SOLUTION We use (3.15) four times:

$$f'(x) = (4 \cdot \frac{3}{2})x^{1/2} = 6x^{1/2}$$

$$f''(x) = (6 \cdot \frac{1}{2})x^{-1/2} = 3x^{-1/2}$$

$$f'''(x) = 3(-\frac{1}{2})x^{-3/2} = -\frac{3}{2}x^{-3/2}$$

$$f^{(4)}(x) = (-\frac{3}{2})(-\frac{3}{2})x^{-5/2} = \frac{9}{4}x^{-5/2}$$

EXERCISES 3.2

Exer. 1–4: (a) Use Definition (3.5) to find $f'(x)$. (b) Find the domain of f' . (c) Find an equation of the tangent line to the graph of f at P . (d) Find the points on the graph at which the tangent line is horizontal.

1 $f(x) = -5x^2 + 8x + 2$; $P(-1, -11)$

2 $f(x) = 3x^2 - 2x - 4$; $P(2, 4)$

3 $f(x) = x^3 + x$; $P(1, 2)$

4 $f(x) = x^3 - 4x$; $P(2, 0)$

Exer. 5–12: (a) Use (3.12)–(3.15) to find $f'(x)$. (b) Find the domain of f' . (c) Find an equation of the tangent line to the graph of f at P . (d) Find the points on the graph at which the tangent line is horizontal.

5 $f(x) = 9x - 2$; $P(3, 25)$

6 $f(x) = -4x + 3$; $P(-2, 11)$

7 $f(x) = 37$; $P(0, 37)$

8 $f(x) = \pi^2$; $P(5, \pi^2)$

- 9 $f(x) = 1/x^3$; $P(2, \frac{1}{8})$
 10 $f(x) = 1/x^4$; $P(1, 1)$
 11 $f(x) = 4x^{1/4}$; $P(81, 12)$
 12 $f(x) = 12x^{1/3}$; $P(-27, -36)$

Exer. 13–16: Find the first three derivatives.

- 13 $f(x) = 3x^6$
 14 $f(x) = 6x^4$
 15 $f(x) = 9\sqrt[3]{x^2}$
 16 $f(x) = 3x^{7/3}$
 17 If $z = 25t^{9/5}$, find $D_t^2 z$.
 18 If $y = 3x + 5$, find $D_x^3 y$.
 19 If $y = -4x + 7$, find $\frac{d^3 y}{dx^3}$.
 20 If $z = 64\sqrt[4]{t^3}$, find $\frac{d^2 z}{dt^2}$.

Exer. 21–22: Is f differentiable on the given interval? Explain.

- 21 $f(x) = 1/x$ (a) $[0, 2]$ (b) $[1, 3)$
 22 $f(x) = \sqrt[3]{x}$ (a) $[-1, 1]$ (b) $[-2, -1]$

Exer. 23–24: Use the graph of f to determine if f is differentiable on the given interval.

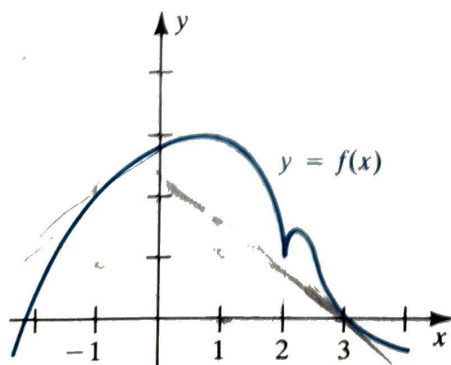
- 23 $f(x) = \sqrt{4-x}$ (a) $[0, 4]$ (b) $[-5, 0]$
 24 $f(x) = \sqrt{4-x^2}$ (a) $[-2, 2]$ (b) $[-1, 1]$

Exer. 25–30: Determine whether f has (a) a vertical tangent line at $(0, 0)$ and (b) a cusp at $(0, 0)$.

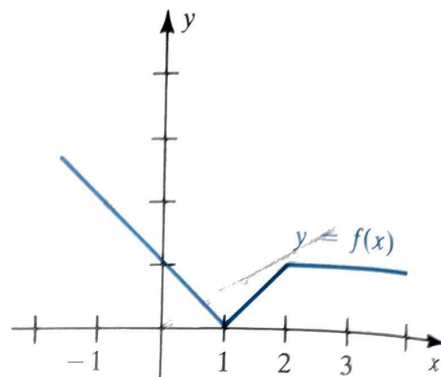
- 25 $f(x) = x^{1/3}$ 26 $f(x) = x^{5/3}$ 27 $f(x) = x^{2/5}$
 28 $f(x) = x^{1/4}$ 29 $f(x) = 5x^{3/2}$ 30 $f(x) = 7x^{4/3}$

Exer. 31–32: Estimate $f'(-1)$, $f'(1)$, $f'(2)$, and $f'(3)$, whenever they exist.

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Exer. 33–36: Use right-hand and left-hand derivatives to prove that f is not differentiable at a .

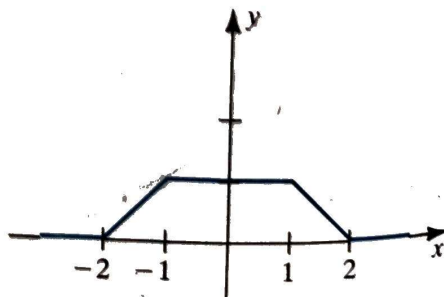
- 33 $f(x) = |x - 5|$; $a = 5$
 34 $f(x) = |x + 2|$; $a = -2$
 35 $f(x) = \llbracket x - 2 \rrbracket$; $a = 2$
 36 $f(x) = \llbracket x \rrbracket - 2$; $a = 2$

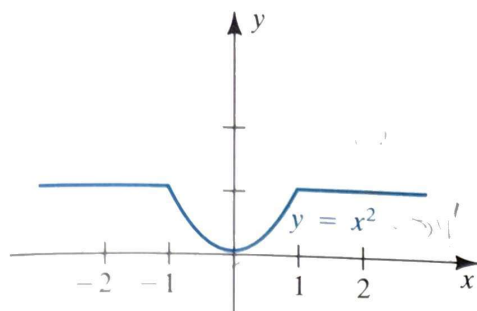
Exer. 37–40: Use the graph of f to find the domain of f' .

- 37 $f(x) = \begin{cases} 2x & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0 \end{cases}$
 38 $f(x) = \begin{cases} 2x - 1 & \text{if } x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$
 39 $f(x) = \begin{cases} -x^2 & \text{if } x < -1 \\ 2x + 3 & \text{if } x \geq -1 \end{cases}$
 40 $f(x) = \begin{cases} x^2 - 2 & \text{if } x < 0 \\ -3 & \text{if } x \geq 0 \end{cases}$

Exer. 41–42: Each figure is the graph of a function f . Sketch the graph of f' and determine where f is not differentiable.

41





Exer. 43–44: Given the position function s of a point P moving on a coordinate line l , find the times at which the velocity is the given value k .

43 $s(t) = 3t^2 - 3$; $k = 4$

44 $s(t) = 4t^3$; $k = 300$

45 The relationship between the temperature F on the Fahrenheit scale and the temperature C on the Celsius scale is given by $C = \frac{5}{9}(F - 32)$. Find the rate of change of F with respect to C .

46 Charles' law for gases states that if the pressure remains constant, then the relationship between the volume V that a gas occupies and its temperature T (in $^{\circ}\text{C}$) is given by $V = V_0(1 + \frac{1}{273}T)$. Find the rate of change of T with respect to V .

47 Show that the rate of change of the volume of a sphere with respect to its radius is numerically equal to the surface area of the sphere.

48 Show that the rate of change of the radius of a circle with respect to its circumference is independent of the size of the circle.

49 An oil spill is increasing such that the surface covered by the spill is always circular. Find the rate at which the area A of the surface is changing with respect to the radius r of the circle at

(a) any value of r (b) $r = 500$ ft

50 A spherical balloon is being inflated. Find the rate at which its volume V is changing with respect to the radius r of the balloon at

(a) any value of r (b) $r = 10$ ft

51 In some applications, function values $f(x)$ may be known only for several values of x near a . In these situations $f'(a)$ is frequently approximated by the formula

$$f'(a) \approx \frac{f(a+h) - f(a-h)}{2h}.$$

(a) Interpret this formula graphically.

(b) Show that $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h} = f'(a)$.

(c) If $f(x) = 1/x^2$, use the approximation formula to estimate $f'(1)$ with $h = 0.1$, 0.01 , and 0.001 .

(d) Find the exact value of $f'(1)$.

52 (a) Use the approximation formula in Exercise 51 to show that if $h \approx 0$, then

$$f''(a) \approx \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}.$$

(b) If $f(x) = 1/x^2$, use part (a) to estimate $f''(1)$ with $h = 0.1$, 0.01 , and 0.001 .

(c) Find the exact value of $f''(1)$.

c Exer. 53–54: Use the following table, which lists the approximate number of feet $s(t)$ that a car travels in t seconds to reach a velocity of 60 mi/hr in 6 seconds.

t	0	1	2	3	4	5	6	7
$s(t)$	0	11.7	42.6	89.1	149.0	220.1	303.7	396.7

53 Use Exercise 51 to approximate the velocity of the car at

(a) $t = 3$ (b) $t = 6$

54 Use Exercise 52 to approximate the rate of change of the velocity of the car with respect to t at

(a) $t = 3$ (b) $t = 6$

c 55 Graph $f(x) = |x^5 - 2x^4 + 3x^3 - x + 1|$ on the interval $[-1, 1]$ and estimate where f is not differentiable.

c 56 Graph $f(x) = x^4 - 3x^3 + 2x - 1$ on the interval $[-1, 3]$ and estimate the x -coordinates of points at which the tangent line is horizontal.

This section contains some general rules that simplify the task of finding derivatives. The rules are stated in terms of the differential operator D_x , where $D_x f(x) = f'(x)$. In the rules, f and g denote differentiable functions,